

Experiment No.	6
Experiment Title	Bagging, Boosting, and Stacked Ensemble Models ¹
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1 Objective

- To understand ensemble learning strategies: Bagging, Boosting, and Stacking.
- To implement Bagging and Boosting classifiers.
- To build a Stacked Ensemble model using multiple base learners.
- To compare ensemble models in terms of accuracy, stability, and generalization.
- To analyze the effect of ensemble methods on bias and variance.

2 Problem Statement

Breast cancer diagnosis from digitized images of a fine needle aspirate (FNA) of a breast mass is a binary classification problem. Given 30 numerical features describing the characteristics of cell nuclei present in the image (e.g., radius, texture, perimeter, area, smoothness), the objective is to predict whether the mass is **Malignant (M)** or **Benign (B)**. In this experiment, three ensemble learning strategies – Bagging, Boosting, and Stacking – are implemented and their predictive performance is compared using standard classification metrics, with the aim of identifying which ensemble strategy generalizes best on unseen data.

3 Dataset Description

Dataset Name	Wisconsin Diagnostic Breast Cancer (WDBC) Dataset
Dataset Source	UCI Machine Learning Repository (https://archive.ics.uci.edu)
Number of Samples	569
Number of Features	30 numerical attributes (10 real-valued features $\times$ 3 statistics – mean, standard error, worst)
Number of Classes	2 (Malignant = M, Benign = B)
Missing Values	0 (no missing values found)
Train-Test Split	80% Training (455 samples) – 20% Testing (114 samples), stratified split

4 Exploratory Data Analysis

Include all relevant visualizations.

- Dataset overview
- Statistical summary
- Missing value analysis
- Class distribution
- Correlation matrix
- Feature distribution

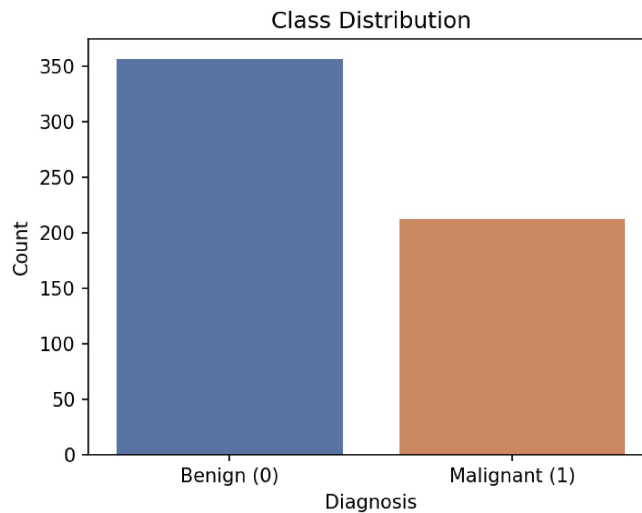


Figure 1: Class distribution of the WDBC dataset (0 = Benign, 1 = Malignant)

Inference:

The dataset contains 357 benign and 212 malignant samples, showing a moderate class imbalance (roughly 63% vs 37%). This imbalance is not severe, but it justifies the use of a stratified train-test split so that both classes are proportionally represented in the training and testing sets. It also motivates evaluating models using precision, recall, and F1-score in addition to accuracy, since accuracy alone can be misleading on imbalanced data. No missing values were found during preprocessing, so no imputation was required.

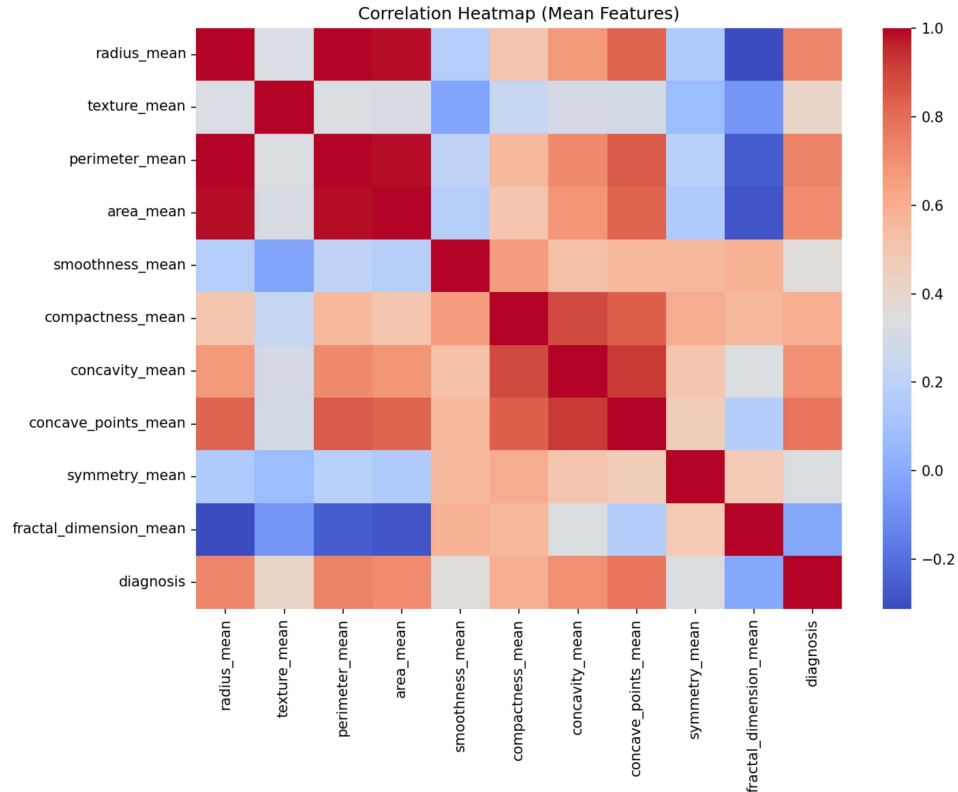


Figure 2: Correlation heatmap of the mean-valued features

Inference:

Several mean features are strongly correlated with one another – for example, **radius_mean**, **perimeter_mean**, and **area_mean** show very high positive correlation, which is expected since perimeter and area are geometrically derived from radius. **concavity_mean**, **concave_points_mean** and **compactness_mean** are also highly correlated with each other and with the target class. This multicollinearity suggests that tree-based ensemble methods (which are relatively robust to correlated features) are well suited to this dataset, while linear meta-learners in the stacked ensemble can still benefit from the redundant signal.

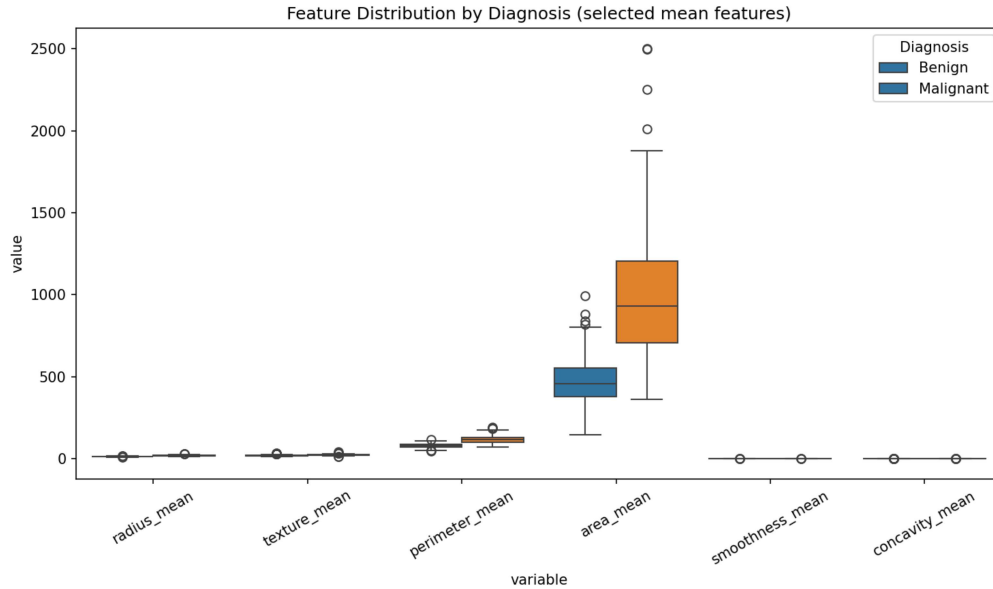


Figure 3: Distribution of selected mean features grouped by diagnosis

Inference:

Malignant samples consistently show higher median values and greater spread for radius, perimeter, area, and concavity compared to benign samples. This separation between the two classes indicates that these features carry strong discriminative power, which explains why relatively simple base learners (Decision Trees, SVM, Naive Bayes) are already able to achieve high accuracy on this dataset, and why ensembling them further improves stability and generalization.

5 Data Preprocessing

- **Missing value handling:** The dataset was checked for missing values using `df.isnull().sum()`; no missing values were found, so no imputation was necessary.
- **Encoding:** The categorical target column `diagnosis` (M/B) was label-encoded into numeric form using `LabelEncoder`, with Malignant mapped to 1 and Benign mapped to 0.
- **Feature scaling:** All 30 numerical features were standardized using `StandardScaler` (zero mean, unit variance) since the features span very different ranges (e.g., area vs. smoothness), which is particularly important for distance/margin-based models such as SVM and Logistic Regression used in the stacked ensemble.
- **Train-test split:** The dataset was split into 80% training and 20% testing sets using `train_test_split` with `stratify=y` to preserve the class ratio, and a fixed `random_state=42` for reproducibility.
- **Feature engineering:** No additional feature engineering was performed; all 30 original features were retained for training.

6 Machine Learning Algorithm

Bagging (Bootstrap Aggregation)

Working principle: Bagging trains multiple copies of a base estimator (here, a Decision Tree) on different bootstrap samples (random samples drawn with replacement) of the training data, and aggregates their predictions by majority voting (for classification).

Mathematical formulation: Given a training set D of size n , B bootstrap samples $D_1, D_2, \dots, D_B$ are drawn (each of size n , with replacement). A model h_b is trained on each D_b , and the final prediction is

$$\hat{y} = \text{mode}\{h_1(x), h_2(x), \dots, h_B(x)\}.$$

Advantages: Reduces variance of high-variance base learners; models can be trained independently (parallelizable); reduces overfitting of a single deep decision tree.

Limitations: Does not reduce bias; if the base learner is already low-variance/high-bias, bagging gives limited improvement; interpretability of individual trees is lost in the ensemble.

Boosting (AdaBoost / Gradient Boosting)

Working principle: Boosting builds an ensemble sequentially, where each new weak learner is trained to correct the errors of the combined ensemble so far. AdaBoost increases the weights of misclassified samples at each iteration, while Gradient Boosting fits each new learner to the residual (negative gradient of the loss) of the current ensemble.

Mathematical formulation (AdaBoost): At iteration t , sample weights w_i are updated as

$$w_i^{(t+1)} = w_i^{(t)} \exp(\alpha_t \cdot \mathbb{I}[y_i \neq h_t(x_i)]), \quad \alpha_t = \frac{1}{2} \ln \left(\frac{1 - \epsilon_t}{\epsilon_t} \right)$$

where ϵ_t is the weighted error of learner h_t . The final prediction is $\hat{y} = \text{sign}(\sum_t \alpha_t h_t(x))$.

Advantages: Reduces bias and can achieve very high accuracy; naturally focuses learning capacity on hard-to-classify samples.

Limitations: Sequential training makes it slower and harder to parallelize; more sensitive to noisy data/outliers (since misclassified noisy points get up-weighted); risk of overfitting if the number of estimators is too large or the learning rate too high.

Stacked Ensemble (Stacking)

Working principle: Stacking trains several heterogeneous base models (SVM, Naive Bayes, Decision Tree) and then trains a meta-learner (Logistic Regression) on the out-of-fold predictions of the base models to learn the optimal way to combine them.

Mathematical formulation: For base learners $h_1, \dots, h_M$, the meta-features for a sample x are $z(x) = [h_1(x), h_2(x), \dots, h_M(x)]$, and the meta-learner g produces the final prediction $\hat{y} = g(z(x))$, where g is trained on out-of-fold predictions to avoid information leakage.

Advantages: Combines diverse, complementary models; often achieves superior performance compared to any single base model; flexible – any classifier can be used as base model or meta-learner.

Limitations: More complex and computationally expensive (requires training M base models plus cross-validation for meta-features); harder to interpret; risk of overfitting the meta-learner if not validated carefully.

7 Experimental Setup

Python Version	3.12
Scikit-Learn Version	1.5
IDE	Jupyter Notebook
Operating System	Ubuntu 24.04 (Linux)
Hardware	Standard CPU (no GPU required)

8 Hyperparameter Selection

Bagging – best parameters from 5-fold GridSearchCV:

Parameter	Value
n_estimators	50
max_samples	1.0
max_features	1.0

Boosting (AdaBoost) – best parameters from 5-fold GridSearchCV:

Parameter	Value
n_estimators	150
learning_rate	1.0

Stacked Ensemble – configuration used:

Parameter	Value
Base Models	SVM, Naive Bayes, Decision Tree
Meta Learner (Final Estimator)	Logistic Regression
Internal CV (for meta-features)	5-fold

9 Results

Test-set results for the best configuration of each ensemble model are reported below (Table 2).

Table 2: Performance Comparison of Ensemble Models on the Test Set

Model	Accuracy (%)	Precision	Recall	F1-score	ROC-AUC
Bagging	96.49	1.0000	0.9048	0.9500	0.9917
Boosting (AdaBoost)	97.37	1.0000	0.9286	0.9630	0.9871
Stacked Ensemble	97.37	1.0000	0.9286	0.9630	0.9974

10 Result Visualization

Include all graphs generated during the experiment.

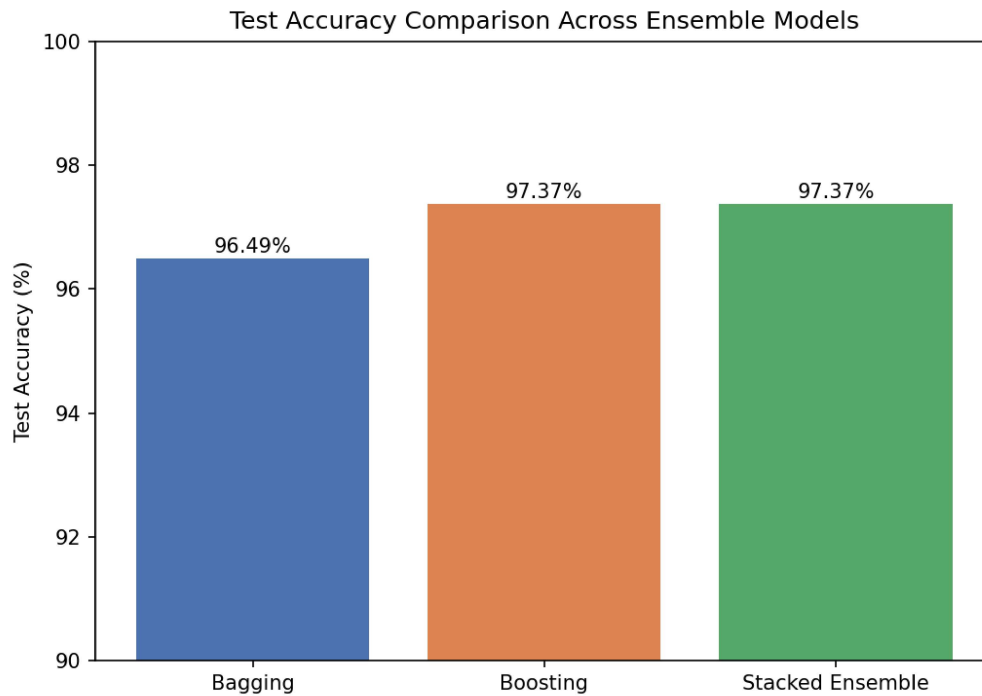


Figure 4: Test accuracy comparison across the three ensemble models

Inference:

Boosting and the Stacked Ensemble achieve the highest test accuracy (97.37%), slightly outperforming Bagging (96.49%). This is consistent with the theory: Boosting reduces bias through sequential error correction, and the Stacked Ensemble benefits from combining diverse base learners. The relatively small gap between all three models (less than 1%) indicates that the WDBC dataset is largely linearly/non-linearly separable and that even a single well-tuned ensemble can perform very well on it.

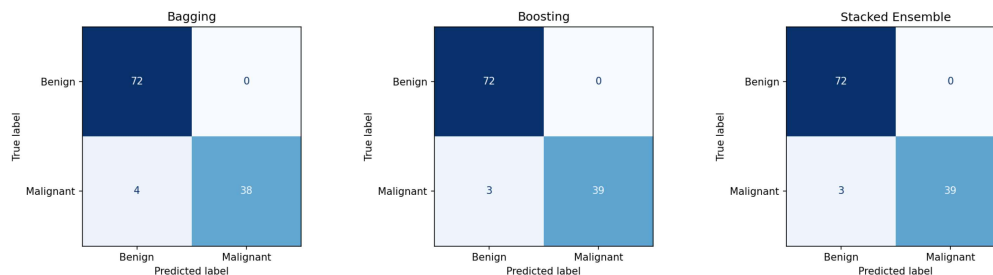


Figure 5: Confusion matrices for Bagging, Boosting, and Stacked Ensemble on the test set

Inference:

All three models achieve perfect precision (no false positives), meaning no benign sample was ever misclassified as malignant. The main source of error across all models is a small number of false negatives (malignant samples predicted as benign) – Bagging has the most (4 out of 42 malignant test cases), while Boosting and Stacking each have 3. In a medical diagnosis context, minimizing false negatives is critical, so the lower false-negative count of Boosting and Stacking is a meaningful practical advantage.

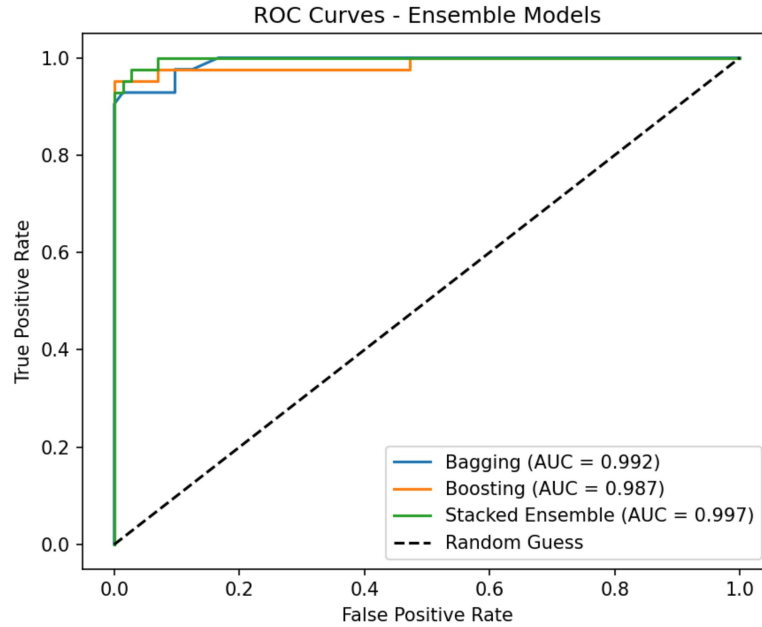


Figure 6: ROC curves and AUC values for the three ensemble models

Inference:

All three models achieve very high AUC values (0.987–0.997), indicating excellent separability between the malignant and benign classes across all decision thresholds. The Stacked Ensemble has the highest AUC (0.9974), suggesting it produces the most reliable probability estimates among the three, even though its default-threshold accuracy ties with Boosting. This makes the Stacked Ensemble particularly attractive if the classification threshold needs to be tuned for clinical sensitivity requirements.

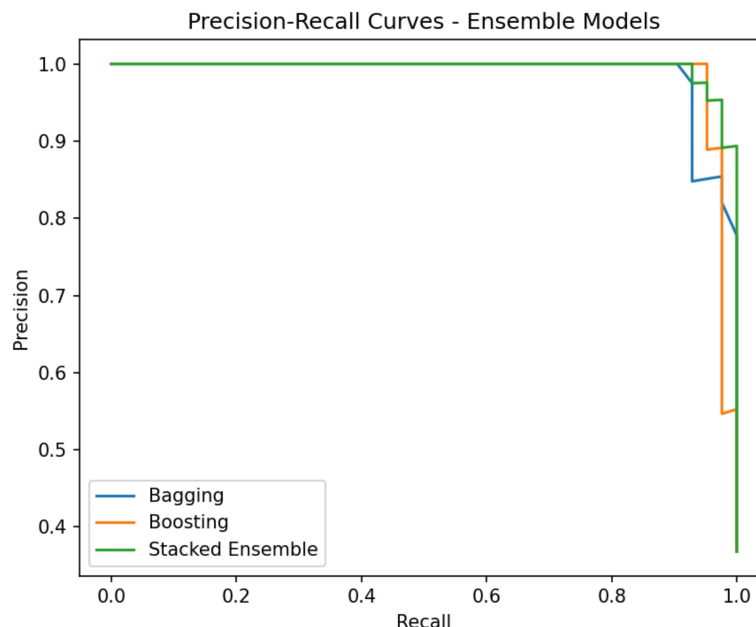


Figure 7: Precision-Recall curves for the three ensemble models

Inference:

All three models maintain high precision across most recall levels, reflecting the earlier observation of zero false positives at the default threshold. The curves stay close to the top-right corner (precision ≈ 1 , recall close to 1) for a large portion of the recall range, which is typical of a well-separated dataset. The Stacked Ensemble’s curve stays marginally higher for longer, consistent with its higher AUC.

11 Discussion

The three ensemble strategies were compared using 5-fold cross-validation for hyperparameter tuning and a held-out 20% test set for final evaluation. Bagging with a Decision Tree base estimator reduced the variance of a single deep tree, achieving 96.49% test accuracy but with a relatively larger train–test accuracy gap (100% train vs. 96.49% test, a 3.5-point gap), indicating some residual overfitting typical of bagged tree ensembles. Boosting (AdaBoost with 150 estimators and a learning rate of 1.0) achieved the same top test accuracy as Stacking (97.37%) by sequentially focusing on hard-to-classify samples, effectively reducing bias; however, it too showed a 100% training accuracy, revealing some overfitting risk from an aggressive learning rate. The Stacked Ensemble (SVM + Naive Bayes + Decision Tree combined via a Logistic Regression meta-learner) achieved the smallest train–test gap (98.90% train vs. 97.37% test, only 1.53 points), the highest ROC-AUC (0.9974), and matched the best test accuracy – suggesting it generalizes the most reliably among the three, thanks to combining diverse, complementary decision boundaries from heterogeneous base learners.

A limitation of this study is the relatively small test set (114 samples), which means differences of a single misclassified sample can shift accuracy by nearly 1%; a larger dataset or repeated cross-validation on the full data would give more statistically ro-

bust comparisons. Potential improvements include using nested cross-validation for hyperparameter tuning (to avoid any optimistic bias in the reported CV scores), trying additional/soft-voting stacking configurations, and applying class-weighting or threshold tuning to further reduce false negatives, which are the most clinically costly errors in this dataset.

12 Conclusion

Bagging, Boosting, and Stacked Ensemble models were successfully implemented and evaluated on the Wisconsin Diagnostic Breast Cancer dataset. Hyperparameters for each model were tuned using 5-fold stratified cross-validation, and all models were compared using accuracy, precision, recall, F1-score, ROC-AUC, confusion matrices, and train-test accuracy gaps. Boosting and the Stacked Ensemble achieved the best test accuracy (97.37%), with the Stacked Ensemble additionally showing the best generalization behavior (smallest train-test gap) and highest ROC-AUC (0.9974). The experiment confirms the theoretical expectations: Bagging primarily reduces variance, Boosting primarily reduces bias, and Stacking leverages the complementary strengths of heterogeneous base learners to achieve the most balanced and robust overall performance.

13 References

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